

Erdős Problem #269

EP-269

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let P be a finite set of primes with $|P| \geq 2$ and let $\{a_1 < a_2 < \dots\} = \{n \in \mathbb{N} : \text{if } p \mid n \text{ then } p \in P\}$. Is the sum $\sum_{n=1}^{\infty} \frac{1}{[a_1, \dots, a_n]}$, where $[a_1, \dots, a_n]$ is the lowest common multiple of $a_1, \dots, a_n$, irrational?

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